oscillating through the spacetime fabric, and if all goes according to plan, LIGO will be the first instrument to tune in.[5]

What makes this all so exciting is that gravitational waves maximize the utility of gravity's two main features: its weakness and its ubiquity. Of all four forces, gravity interacts with matter most feebly. This implies that gravitational waves can pass through material that's opaque to light, giving access to astrophysical realms previously hidden. What's more, because *everything* is subject to gravity (whereas, for example, the electromagnetic force only affects objects carrying an electric charge), everything has the capacity to generate gravitational waves and hence produce an observable signature. LIGO thereby marks a significant turning point in the way we examine the cosmos.

There was a time when all we could do was raise our eyes and gaze skyward. In the seventeenth century, Hans Lippershey and Galileo Galilei changed that; with the aid of the telescope, the grand vista of the cosmos came within humanity's purview. But in time, we realized that visible light represented a narrow band of electromagnetic waves. In the twentieth century, with the aid of infrared, radio, X-ray, and gamma ray telescopes, the cosmos opened up to us anew, revealing wonders invisible in the wavelengths of light that our eyes have evolved to see. Now, in the twenty-first century, we are opening up the heavens once again. With LIGO and its subsequent improvements,[39] we will view the cosmos in a completely new way. Rather than using electromagnetic waves, we will use gravitational waves; rather than using the electromagnetic force, we will use the gravitational force.

To appreciate how revolutionary this new technology may be, imagine a world on which alien scientists were just now discovering how to detect electromagnetic waves—light—and think about how their view of the universe would, in short order, profoundly change. We are on the cusp of our first detection of gravitational waves and so may well be in a similar position. For millennia we have looked into the cosmos; now it's as if, for the first time in human history, we will listen to it.

The Hunt for Extra Dimensions